

Tide retrospective: *general- h partition derivatives and the formal $Z''(0), Z'''(0)$*

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Abstract

The previous tide proved the iterated-derivative *step* for the anharmonic perturbed partition family $G_n(h) = \int (-tx)^n e^{-t(L+hx)}$ only at $h = 0$, which gives the derivative of each family member but does not formally identify the higher derivatives of $Z = G_0$. Following GPT-5.5 Pro’s recommendation, this tide proves the **local-in- h** step — for every n and $|h| < 1$, $\frac{d}{dh}G_n(h) = G_{n+1}(h)$ — and chains it through `iteratedDeriv` to land the genuine identities

$$Z''(0) = \int (tx)^2 e^{-tL}, \quad Z'''(0) = \int -(tx)^3 e^{-tL}.$$

New file `AnharmonicPartitionDerivGeneralH.lean`; commit `a7886cd` on branch `tide/anharmonic-partition-deriv-general-h`; full library builds (8304 jobs), no `sorry/axiom/native_decide`, no lint warnings.

1 Setting

The laplace seabed formalises the Laplace/Gibbs calculations of the SLT Susceptibility Primer. The prior tide¹ introduced the weighted-partition family $G_n(h) = \int (-tx)^n e^{-t(L+hx)}$ and proved $G'_n(0) = G_{n+1}(0)$. In its post-merge consult, GPT-5.5 Pro flagged the gap: proving the step only at $h = 0$ shows that each G_n has $G_{n+1}(0)$ as its derivative at 0, but to call $G_2(0)$ “ $Z''(0)$ ” one must know that $\text{deriv } Z = G_1$ on a neighbourhood of 0 so the second derivative can be taken. The fix it recommended — prove the local-in- h step, then chain — is this tide.

2 The thing formalised

For $\lambda, \gamma > 0$, $\alpha^2 < 3\lambda\gamma$, $t > 0$, every $n \in \mathbb{N}$ and every h with $|h| < 1$:
 $\text{HasDerivAt } G_n (G_{n+1}(h)) \ h$. Consequently $\text{deriv}(G_n)(h) = G_{n+1}(h)$ for $|h| < 1$, $\text{iteratedDeriv } k \ G_0 \ h = G_k(h)$, and in particular $Z''(0) = \int (tx)^2 e^{-tL}$, $Z'''(0) = \int -(tx)^3 e^{-tL}$.

The headline Lean signatures:

```
theorem weightedPartition_hasDerivAt (n : ℕ) ... {h₀ : ℝ} (hh₀ : |h₀| < 1) :
  HasDerivAt (weightedPartition lam alpha gamma t n)
    (weightedPartition lam alpha gamma t (n + 1) h₀) h₀
```

¹the higher-derivative step tide.

```

theorem iteratedDeriv_two_partition_zero ... :
  iteratedDeriv 2 (weightedPartition lam alpha gamma t 0) 0
    = ∫ x, (t * x) ^ 2 *
      Real.exp (-(t * anharmonicPotential lam alpha gamma x))

```

3 Proof strategy

The step. The dominated-differentiation argument is the prior tide’s, with the centre moved from 0 to an arbitrary h_0 with $|h_0| < 1$. Two things change. First, the neighbourhood: `ball01` is open and contains h_0 , hence is a neighbourhood of h_0 , so the same ball and the same $|h| < 1$ bound serve at any interior centre. Second, the dominated-differentiation lemma needs integrability of the family *at the centre* h_0 ; with $h_0 \neq 0$ the $+0 \cdot x$ simplification of the $h = 0$ case is gone, so this needs the bare-exponential bound $e^{-t(L+h_0x)} \leq e^{t/2c} e^{-(t/2)L}$ ($|h_0| < 1$), which I extracted as `anharmonic_perturbed_exp_le` (the exponent half of the existing weighted pointwise bound; `amgm_t_abs_x` un-privated to support it).

The chaining. With $\text{deriv}(G_n)(h) = G_{n+1}(h)$ on `ball01` in hand, an induction on k gives $\text{iteratedDeriv } k \, G_0 \, h = G_k(h)$ for $|h| < 1$: at the successor, the inductive hypothesis holds on the whole open ball, hence `iteratedDeriv k G_0` agrees with G_k on a neighbourhood of h , so `Filter.EventuallyEq.deriv_eq` turns $\text{iteratedDeriv } (k+1) \, G_0 \, h = \text{deriv}(\text{iteratedDeriv } k \, G_0) \, h$ into $\text{deriv}(G_k)(h) = G_{k+1}(h)$. The $k = 2, 3$ specialisations at $h = 0$, with the integrand simplified by `ring` (treating the exponential as an atom), are the headline $Z''(0)$, $Z'''(0)$.

4 Roadblocks and resolutions

The chaining went in cleanly once phrased as “agree on the open ball” rather than “agree at a point”: it is the open-ball membership that supplies the `EventuallyEq` the derivative-rewrite needs, and stating the induction hypothesis as $\forall h, |h| < 1 \rightarrow \dots$ rather than a single `EventuallyEq` made the successor step’s neighbourhood argument a two-line `Filter.eventuallyEq_of_mem`.

The one build failure was cosmetic: the final integrand-matching used `congr 1; ring` on $-(tx)^k = \pm(tx)^k$, but `congr` tries to peel the power and strands an impossible base equality. Rewriting the exponential factor first and then closing with a single `ring` (the exponential is an opaque atom, the rest is a polynomial identity) fixed both the $k = 2$ and $k = 3$ cases.

5 What was Mathlib, what was new

Mathlib. The dominated-differentiation lemma²; `HasDerivAt.deriv`; `iteratedDeriv_zero`, `iteratedDeriv_succ`; `Filter.EventuallyEq.deriv_eq`, `Filter.eventuallyEq_of_mem`; `Metric.isOpen_ball.mem_nhds`.

Seabed (reused). The general $|x|^m$ integrability witness, the two public pointwise helpers, and `anharmonic_coercive` — all from the two earlier anharmonic tides this day; `amgm_t_abs_x` (un-privated).

New. The bare-exp domination bound; integrability at a nonzero centre; the general- h step; the `iteratedDeriv` induction; and the two genuine partition-derivative identities. About 250 lines.

²`hasDerivAt_integral_of_dominated_loc_of_deriv_le`.

6 Lessons

- **A post-merge consult is still worth acting on.** GPT’s “prove the local-in- h version” arrived after the $h = 0$ tide had merged; treating it as the next tide rather than ignoring it turned a derivative-of-a-family-member into the genuinely useful $Z''(0)$, $Z'''(0)$.
- **For iteratedDeriv chaining, carry the hypothesis on an open set, not a point.** The neighbourhood agreement that `EventuallyEq.deriv_eq` consumes falls straight out of open-ball membership; phrasing the induction over $|h| < 1$ avoids any bespoke `EventuallyEq` bookkeeping.
- **ring with an opaque exp atom beats congr for integrand matching.** `congr` peels powers and strands false base goals; rewriting the exponential then `ring` closes signed-power identities in one step.

7 Follow-ups

- **Fourth-cumulant / flow-equation identity.** With $Z(0) > 0$ and $Z', Z'', Z''', Z^{(4)}$ at 0 all available (the last is the $k = 4$ specialisation), assemble the connected fourth cumulant $\kappa_4 = \partial_h^4 \log Z|_0$. This is the motivating application and is now fully unblocked.
- **Moment normalisation.** $Z^{(n)}(0)/Z(0) = \langle (-tx)^n \rangle_0$ packages the derivatives as raw moments under the unperturbed Gibbs measure.
- **Promote `amgm_t_abs_x` to `lean-common`.** Now load-bearing across three anharmonic tides and genuinely generic.